

National University of Singapore

CS2109S—Introduction to AI and Machine Learning

Semester 2, 2021/2022

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS

1. Write down your **Student Number** on the answer sheet and shade completely the corresponding bubbles in the grid for each digit or letter. **DO NOT WRITE YOUR NAME!**
2. The assessment paper contains **SIX (6) questions** and comprises **TEN (10) pages** including this cover page.
3. Weightage of questions is given in square brackets. The maximum attainable score is 100.
4. This is a **OPEN-SHEET** assessment. You are allowed one A4-sized double-sided cheat-sheet.
5. You are allowed to bring a calculator, but it cannot have any form of external communication capability, i.e. not Wifi- or 4G-enabled. Mobile phones and tablets are not allowed.
6. All questions must be answered in the space provided on the answer sheet; no extra sheets will be accepted as answers.
7. You are allowed to write with pencils, as long as it is legible.
8. **Marks may be deducted** for unrecognisable handwriting and/or for not shading the student number properly.
9. You must submit only the **ANSWER SHEET** and no other documents. The question set may be used as scratch paper.
10. An excerpt of the question may be provided in the answer sheet to aid you in answering in the correct box, where applicable. It is not the exact question. You should still refer to the original question in this question booklet.

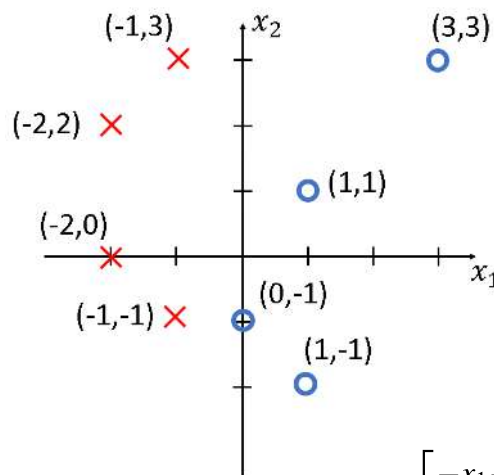
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It may be used as scratch paper.

Question 1: Short Questions [14 marks]

A. Explain what you understand is the “kernel trick” and describe situations where we might want to use it. [3 marks]

B. Consider the 8 training data points in the figure below. There are 2 classes: circles and crosses. Suppose we apply an SVM to classify the data. Circle the support vector(s). [3 marks]



C. [Alternative PCA Formulation] Suppose that $\mathbf{X} = \begin{bmatrix} -x_1 - \\ -x_2 - \\ \vdots \\ -x_m - \end{bmatrix}$ is an array of the training

samples in row format, then the following is an implementation of the SVD-based approach to PCA discussed in lecture to transform the samples x_i from n degrees to k degrees ($k < n$, by performing SVD on the “centred” covariance matrix $\mathbf{X}^T \mathbf{X}$:

```
sigma= X.T @ X
U, s, VT = svd(sigma)
Ur = U[:, :k]
Z = X @ Ur
Xapprox = Z @ Ur.T
```

It turns out that there is an alternative formulation where we can perform SVD *directly* on $\mathbf{X}$ to yield the *same(!)* results:

```
U, s, VT = svd(X)
vectors = VT[:k]
Z = X @ vectors.T
Xapprox2 = Z @ vectors
```

What do you think is the practical difference between these 2 implementations and why do you think we would prefer one formulation over the other. You are expected to reason about this from first principles. [4 marks]

D. You apply Gradient Descent to a ML-problem and you found that the training takes a long time. Suggest 2 ways that might be able to help you reduce the training time without sacrificing too much accuracy. [4 marks]

Question 2: COVID-Rho [31 marks]

After the latest wave of the Omicron variant subsided, scientists were horrified to discovered a new variant, for which existing ART tests no longer work! They call this the *Rho* variant. The symptoms of the new variant, while similar to previous variants are not entirely the same. Since existing ART tests no longer work, the government needs a new way to determine if the sick people who come to A&E are likely to have contracted the new variant.

Since you have taken CS2109S, you were approached to construct a decision tree classifier. Unfortunately, there is not much data available since and all you have are the data for 12 patients shown in the table below. There are 3 possible outcomes for the current detailed lab analysis: Positive (for Rho), Negative (for Rho), and Other (some other non-Rho COVID variant). The breaking order on constructing tree is No Taste > Tiredness > Cough > Fever.

Patient	Fever?	Cough?	Tiredness?	No Taste?	Rho?
A	Yes	No	Yes	No	Negative
B	No	Yes	Yes	Yes	Positive
C	Yes	Yes	No	Yes	Other
D	Yes	No	No	Yes	Other
E	Yes	No	Yes	Yes	Positive
F	Yes	Yes	No	No	Negative
G	No	Yes	Yes	No	Positive
H	No	Yes	Yes	No	Negative
I	Yes	No	Yes	Yes	Positive
K	No	Yes	No	Yes	Negative
L	No	No	No	No	Negative
M	No	No	Yes	Yes	Positive

A summary of count for these results is given in the following table for your convenience:

Summary	Fever?	Cough?	Tiredness?	No Taste?
Yes/Positive	2	2	5	4
Yes/Negative	2	3	2	1
Yes/Other	2	1	0	2
No/Positive	3	3	0	1
No/Negative	3	2	3	4
No/Other	0	1	2	0
Total	12	12	12	12

The information content (entropy) for a given probability distribution p_i , for $i = 1, \dots, n$ is given by:

$$-\sum_{i=1}^n p_i \log_2(p_i)$$

A. [Warm Up] What is the entropy of the COVID outcomes?

Hint: remember the log is in base 2!

[3 marks]

B. Construct a decision tree using information gain to predict whether a given patient is likely to have COVID-Rho. [15 marks]

We break the tie in favour of Cough over Fever.

C. There is likely some error in the initial data leading to overfitting. Derive a pruned decision tree that has at least 2 training data points for each leaf. [4 marks]

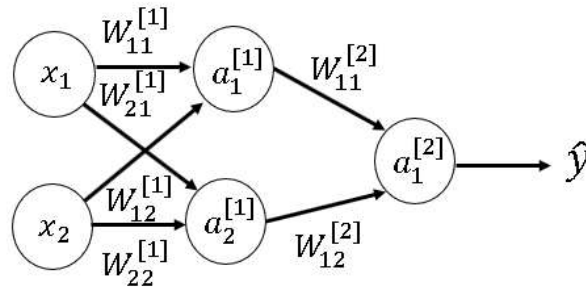
D. Given the following test data, what is the precision, recall and F1 value for the pruned decision tree in Part(C). You can consider “Other” to be “Negative.” [6 marks]

Patient	Fever?	Cough?	Tiredness?	No Taste?	Ground Truth
1	Yes	Yes	No	No	Negative
2	No	No	No	No	Negative
3	Yes	Yes	No	Yes	Other
4	Yes	No	No	Yes	Positive
5	No	Yes	No	Yes	Other
6	Yes	No	Yes	No	Negative
7	No	Yes	Yes	No	Negative
8	No	Yes	Yes	No	Negative
9	No	Yes	Yes	Yes	Positive
10	Yes	No	Yes	Yes	Positive

E. Suppose you are told that COVID-Rho is very very deadly compared to the other variants, which would you try to maximize: precision or recall? Explain. [3 marks]

Question 3: BackProp Calculations [20 marks]

Consider the following simple 2-layer MLP network with 1 hidden layer and no bias terms:



Following the notation used in lecture:

$$\begin{aligned} \mathbf{f}^{[1]} &= (\mathbf{W}^{[1]})^T \mathbf{x} \\ \mathbf{a}^{[1]} &= g^{[1]}(\mathbf{f}^{[1]}) \\ \mathbf{f}^{[2]} &= (\mathbf{W}^{[2]})^T \mathbf{a}^{[1]} \\ \hat{y} &= g^{[2]}(\mathbf{f}^{[2]}) \end{aligned}$$

All the activation functions are sigmoids given by:

$$g^{[1]}(x) = g^{[2]}(x) = \sigma(x) = \frac{1}{1 + e^{-x}}$$

For reference,

$$\begin{aligned} \sigma'(x) &= \sigma(x)(1 - \sigma(x)) \\ \mathbf{W}^{[1]} &= \begin{bmatrix} W_{11}^{[1]} & W_{21}^{[1]} \\ W_{12}^{[1]} & W_{22}^{[1]} \end{bmatrix} \\ \mathbf{W}^{[2]} &= \begin{bmatrix} W_{11}^{[2]} \\ W_{12}^{[2]} \end{bmatrix} \end{aligned}$$

Recall that our goal is to compute $\frac{\partial \hat{y}}{\partial \mathbf{W}}$ so that we can use it for Gradient Descent. Also, where L is the maximum layer of an MLP (here, $L = 2$), we define $\delta^{[l]}$ as follows:

$$\delta^{[l]} = \frac{\partial \hat{y}}{\partial \mathbf{f}^{[l]}} \quad (1)$$

- A.** [Warm Up: Forward Propagation] Express $\hat{y}$ in terms of $\mathbf{x}$, $\mathbf{W}^{[1]}$, $\mathbf{W}^{[2]}$ and σ . [2 marks]
- B.** Express $\delta^{[2]}$ in terms of $\mathbf{f}^{[i]}$ and σ , where $i = 1$ or 2 . [2 marks]
- C.** Express $\frac{\partial \hat{y}}{\partial \mathbf{W}^{[2]}} = \frac{\partial \mathbf{f}^{[2]}}{\partial \mathbf{W}^{[2]}} \frac{\partial \hat{y}}{\partial \mathbf{f}^{[2]}}$ in terms of $\mathbf{a}^{[i]}$, and σ , where $i = 1$ or 2 . [2 marks]
- D.** Express $\frac{\partial \hat{y}}{\partial \mathbf{W}^{[2]}}$ in terms of $\delta^{[i]}$ and $\mathbf{a}^{[i]}$, where $i = 1$ or 2 . [2 marks]
- E.** Derive an expression for $\delta^{[1]} = \left(\frac{\partial \mathbf{a}^{[1]}}{\partial \mathbf{f}^{[1]}}\right)^T \frac{\partial \mathbf{f}^{[2]}}{\partial \mathbf{a}^{[1]}} \frac{\partial \hat{y}}{\partial \mathbf{f}^{[2]}}$ in terms of $\delta^{[2]}$ from Equation (1). [3 marks]
- F.** Express $\frac{\partial \hat{y}}{\partial \mathbf{W}^{[1]}} = \frac{\partial \mathbf{f}^{[1]}}{\partial \mathbf{W}^{[1]}} \left(\frac{\partial \hat{y}}{\partial \mathbf{f}^{[1]}}\right)^T$ in terms of $\mathbf{x}^{[0]}$, and $\delta^{[i]}$, where $i = 1, 2$. [3 marks]
- G.** Given $\mathbf{x}^{[0]} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{W}^{[1]} = \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}$ and $\mathbf{W}^{[2]} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, compute $\frac{\partial \hat{y}}{\partial \mathbf{W}^{[1]}}$ and $\frac{\partial \hat{y}}{\partial \mathbf{W}^{[2]}}$ (to 3 significant digits). [6 marks]

Question 4: Unsupervised Learning [12 marks]

In this problem, you and your team are conducting a survey on the population on behalf of the government. The survey consists of 1000 questions, ranging from simple yes/no questions, to approval ratings from 0-10. Your team has managed to gather over one million responses.

- A.** Your team now wants to identify if there are natural clusters found within the population, and intend to do so by performing k-means on the raw data. Suggest a method to go about identifying a good value of k to select. [4 marks]
- B.** In order to initialize the centroids to run k-means, Ben Bitdiddle offers you a strategy to initialize the centroids. Randomly assign every sample to one of the k clusters, and compute the k centroids of this random configuration to use as the initial centroids. Explain why this might not be a good method of random initialization of centroids. [4 marks]
- C.** You have run k-means on your dataset but realised that the clusters do not make much sense. What could have caused this issue, and how can you resolve it? [4 marks]

Question 5: Convolutional Neural Networks [20 marks]

You are given the following rectangular image $\mathbf{x} \in \mathbb{R}^{3 \times 6}$:

A	B	C	D	E	F
G	H	I	J	K	L
M	N	O	P	Q	R

A. If we use k 3×6 kernels, equal to the dimensions of the input image, is this simply a fully connected layer with k output neurons? Explain. [2 marks]

B. You are provided a single 3×3 kernel as shown below. Assume we apply no padding and a stride of 2×2 . What is the value of the bottom-right output pixel in terms of the variables $\in \{A, B, \dots, Q, R\}$ after applying this kernel. You should not flip the kernel. [2 marks]

1	0	-1
1	0	-1
1	0	-1

C. Using the same 3×3 kernel as above, assume we apply a zero-padding of 1 and a stride of 1×1 . What is the value of the bottom-right output pixel now? You should not flip the kernel. [2 marks]

D. Your class has been tasked to solve a image binary classification problem. Your friend Ben Bitdiddle claims that his deep neural network consisting of multiple fully connected layers will be able to solve this image classification problem.

You disagree and feel that we should use convolutional layers early in the network in order to solve the problem. Who is correct? Explain. [3 marks]

E. You design a CNN binary classifier that takes in a 3-channel colour 28x28 image as the input. The model is described as a sequence of layers:

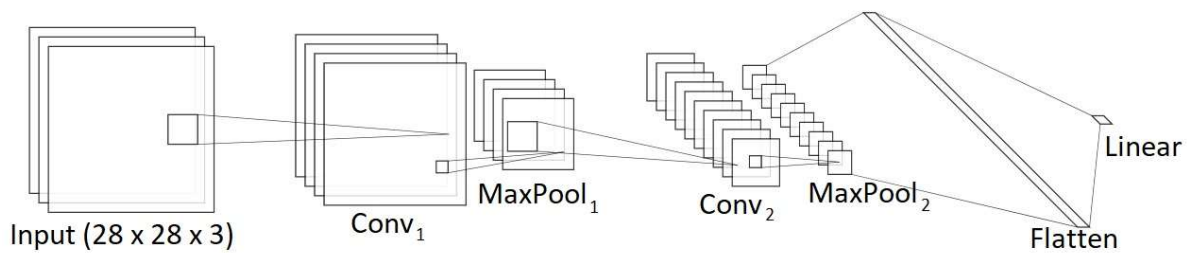
$$\text{Conv}_1(4, (5, 5)) \rightarrow \text{MaxPool}_1 \rightarrow \text{Conv}_2(10, (5, 5)) \rightarrow \text{MaxPool}_2 \rightarrow \text{Flatten} \rightarrow \text{Linear}(x, 1)$$

Conv_1 applies a 5×5 kernel with 0 padding and stride 1, and has 4 output channels.

Conv_2 applies a 5×5 kernel with 0 padding and stride 1, and has 10 output channels.

Both MaxPool layers apply a 2×2 kernel with a stride equal to the kernel dimensions (2×2).

The Linear layer is fully connected from x nodes (to be determined) in the flattened layer to 1 output node for binary classification.



State the dimensions of each layer, and the number of trainable parameters. [6 marks]

F. For the image binary classification problem, you have been provided 1,000 labelled images, 900 of which are labelled class A, and the remaining are labelled class B. Describe in a step-by-step format, how you will set up your experiment to train and evaluate the performance of your classifier. [5 marks]

Question 6: Did you see the animal? [3 marks]

In the “Invisible Gorilla” experiment of 1999, volunteers were asked to watch a video where two groups of people—some dressed in white, some in black—are passing basketballs around. The volunteers were asked to count the passes among players dressed in white while ignoring the passes of those in black. About half of watchers missed a person in a gorilla suit walking in and out of the scene thumping its chest.

In CS2109S, we attempted to verify the results of this study. During one of the lectures, Prof Ben transformed into an animal. What animal was that?

If you cannot remember the animal, you can also try to tell us a joke. If it’s funny enough, maybe you might get some points too. [3 marks]

Appendix

The following is one of the algorithms that was introduced in class that is reproduced here for your reference.

```

function DECISION-TREE-LEARNING(examples, attributes, default) returns a
decision tree

  inputs: examples, set of examples
           attributes, set of attributes
           default, default value for the goal predicate

  if examples is empty then return default
  else if all examples have the same classification then return the classification
  else if attributes is empty then return MAJORITY-VALUE(examples)
  else
    best ← CHOOSE-ATTRIBUTE(attributes, examples)
    tree ← a new decision tree with root test best
    for each value  $v_i$  of best do
      examplesi ← {elements of examples with best =  $v_i$ }
      subtree ← DECISION-TREE-LEARNING(examplesi, attributes − best,
                                         MAJORITY-VALUE(examples))
      add a branch to tree with label  $v_i$  and subtree subtree
    end
  return tree

```

— HAPPY HOLIDAYS —

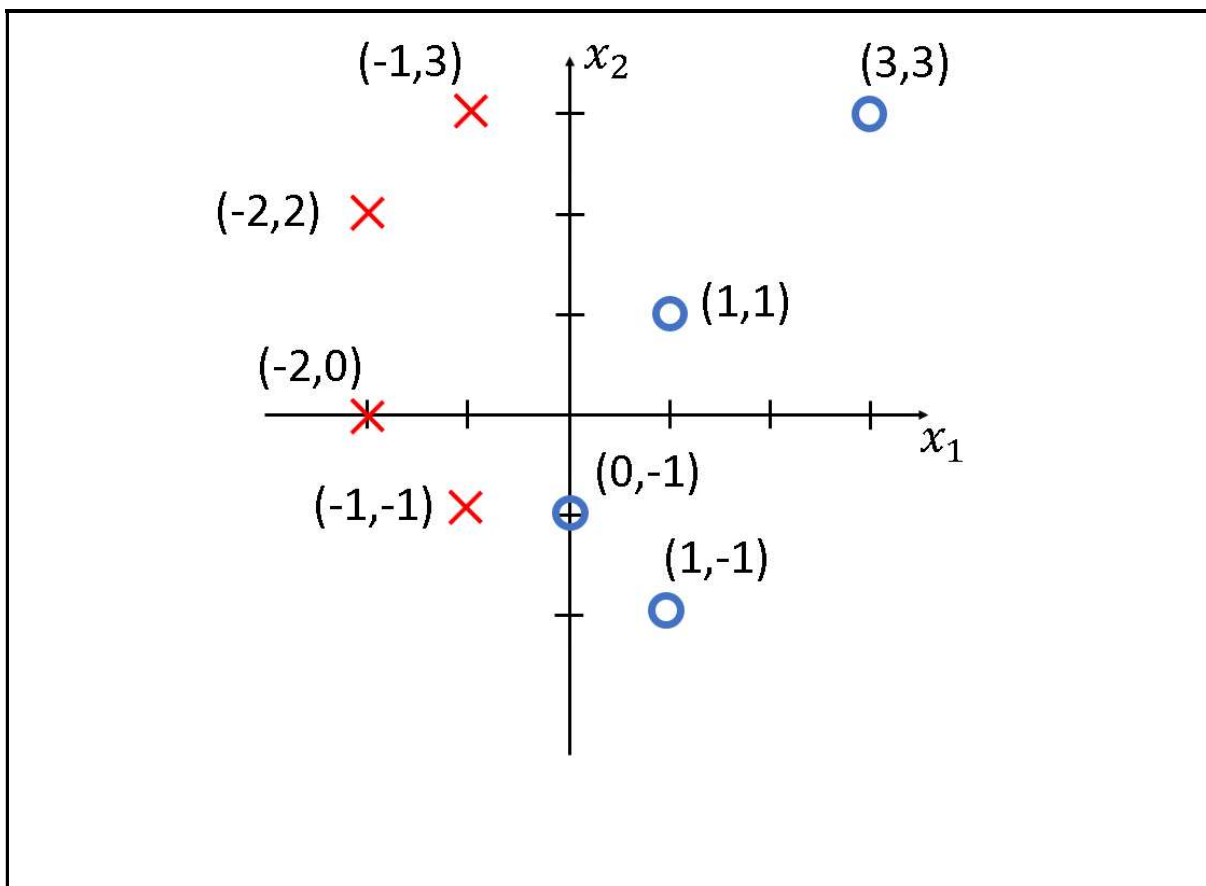
Time allowed: 2 hours

Question 1A What is the “kernel trick”?

[3 marks]

Question 1B Circle the support vector(s).

[3 marks]



Question 1C Alternative PCA Formulation.

[4 marks]

Question 1D Speeding up Gradient Descent.

[4 marks]

Question 2A Entropy of the COVID outcomes

[3 marks]

Question 2B Decision tree to decide if patient has COVID-Rho.

[15 marks]

Question 2B Decision tree to decide if patient has COVID-Rho (continued). [15 marks]

Question 2B Decision tree to decide if patient has COVID-Rho (continued). [15 marks]

Question 2C Pruned decision tree (min-leaf ≥ 2).

[4 marks]

Question 2D Precision, recall and F1 value.

[6 marks]

Question 2E Choice between precision and recall.

[3 marks]

Question 3A Warm Up: Forward Propagation

[2 marks]

Question 3B Find $\delta^{[2]}$.

[2 marks]

Question 3C Find $\frac{\partial \hat{y}}{\partial \mathbf{w}^{[2]}}$.

[2 marks]

Question 3D Find $\frac{\partial \hat{y}}{\partial \mathbf{w}^{[2]}}$ (again).

[2 marks]

Question 3E Find $\delta^{[1]}$.

[3 marks]

Question 3F Find $\frac{\partial \hat{y}}{\partial \mathbf{w}^{[1]}}$.

[3 marks]

Question 3G Compute $\frac{\partial \hat{y}}{\partial \mathbf{w}^{[1]}}$ and $\frac{\partial \hat{y}}{\partial \mathbf{w}^{[2]}}$ (to 3 significant digits). [6 marks]

Question 4A Suggest a method to select a good value of k . [4 marks]

Question 4B Explain why this random initialization method might not be good. [4 marks]

Question 4C What is causing poor k-means performance and how to resolve? [4 marks]

Question 5A Does a full 3×6 kernel behave as a fully connected layer? Explain. [2 marks]

Question 5B Value of the bottom-right output pixel? Padding 0, stride 2. [2 marks]

Question 5C Value of the bottom-right output pixel? Padding 1, stride 1. [2 marks]

Question 5D Should we use CNN or not? Explain. [3 marks]

Question 5E CNN classifier dimensions and number of parameters. [6 marks]

Layer	Layer dimension	Number of trainable parameters
Input	$28 \times 28 \times 3$	0
Conv ₁		
MaxPool ₁		
Conv ₂		
MaxPool ₂		
Linear	$\times 1$	

Question 5F Describe your CNN experiment setup. [5 marks]

Question 6 Did you see the animal? [3 marks]

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It may be used as scratch paper.

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